An exact obstruction to pairwise orthogonality
MathDB 376318

Research note

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Task 1. Resolve the original conjecture

Kuijper's Conjecture 2.25 [1, p. 26] asserts that the multivariate Hermitian Jacobi polynomials in Definition 2.23 are orthogonal whenever their indices are non-equivalent. That definition fixes a multivariate Schur leading term and subtracts strictly lower terms in the specified partial order; it does not impose orthogonality between incomparable indices. We disprove the conjecture, including its allowance for arbitrary choices within repeated representation types.

Set $m = 3$, $k = 6$, and

$$\alpha_j = \beta_j = \tfrac{3}{2} \quad (1 \le j \le 6), \qquad \tau_1 = \cdots = \tau_6 = (1).$$

Thus $X_1, \dots, X_6$ are independent and uniformly distributed on $0 < X < I_3$ in the real vector space of complex Hermitian matrices. These parameters lie strictly inside the source's admissible range. All inner products below are integration against this probability law. The two final partition labels are

$$\lambda = (4, 1, 1), \qquad \mu = (3, 3).$$

They are distinct and incomparable in dominance order.

Theorem 1. *For these parameters, the spaces spanned by the source's Jacobi polynomials with final labels λ and μ are not orthogonal. This holds for every choice of the multivariate Schur bases within their multiplicity spaces. Explicit real polynomials p_λ and p_μ in those respective spans satisfy*

$$\boxed{\langle p_\lambda, p_\mu \rangle = -\frac{395}{12107408}\left(\frac{3}{70}\right)^6 = -\frac{57591}{284884888758400000} < 0.}$$

Consequently at least one pair of the original Jacobi basis polynomials with these non-equivalent partition labels has nonzero inner product, contradicting Conjecture 2.25.

1. Matrix moments

Write $Z = X - I_3/2$. Unitary invariance and invariance under $X \mapsto I - X$ give $\mathbb{E}X = I_3/2$ and

$$\mathbb{E}(X_{ab}X_{cd}) = A\delta_{ab}\delta_{cd} + B\delta_{ad}\delta_{bc}.$$

The unordered eigenvalues have density proportional to $\Delta(x)^2 = \prod_{i<j}(x_i - x_j)^2$ on $[0,1]^3$. Direct integration of this polynomial gives

$q(x)$	1	$\sum_i x_i$	$\sum_i x_i^2$	$(\sum_i x_i)^2$
$\int_{[0,1]^3} q(x)\Delta(x)^2\,dx$	$1/360$	$1/240$	$13/4200$	$9/1400.$

These identities can be checked simply by integrating each monomial, $\int_{[0,1]^3} x_1^i x_2^j x_3^\ell \, dx = ((i+1)(j+1)(\ell+1))^{-1}$. Hence $3A + 9B = 39/35$ and $9A + 3B = 81/35$, so that $A = 17/70$, $B = 3/70$. In particular,

$$\mathbb{E}(Z_{ab} Z_{cd}) = a\delta_{ad}\delta_{bc} + b\delta_{ab}\delta_{cd}, \qquad a = \frac{3}{70}, \quad b = -\frac{1}{140}, \quad b/a = -\frac{1}{6}. \tag{1}$$

This also follows from stationarity of the source's equation (8): its degree-two moment equations are $12A + 2B = 3$, $2A + 12B = 1$.

2. Concrete polynomials and their partition blocks

Let $V = \mathbb{C}^3$. Denote by T_σ the permutation of tensor factors on $V^{\otimes 6}$ corresponding to $\sigma \in S_6$. For a partition $\eta \vdash 6$ of length at most three, let f^η be the dimension of its symmetric-group representation and let $D_\eta = \dim S_\eta(V)$. The central orthogonal projector onto the η-isotypic summand is

$$E_\eta = \frac{f^\eta}{720} \sum_{\sigma \in S_6} \chi^\eta(\sigma) T_\sigma. \tag{2}$$

Define the explicit multilinear polynomials

$$q_\eta(X_1, \dots, X_6) = D_\eta^{-1} \operatorname{Tr}\big(E_\eta(X_1 \otimes \cdots \otimes X_6)\big),$$
$$f_\eta(X_1, \dots, X_6) = D_\eta^{-1} \operatorname{Tr}\big(E_\eta(Z_1 \otimes \cdots \otimes Z_6)\big).$$

They are real on Hermitian arguments and invariant under simultaneous unitary conjugation and under all permutations of the six arguments. Formula (2) makes them finite, fully specified polynomials.

Here is the identification with the source's multiplicity spaces. Schur–Weyl decomposition gives

$$V^{\otimes 6} = \bigoplus_{\eta \vdash 6,\, \ell(\eta) \le 3} S_\eta(V) \otimes F_\eta, \qquad \operatorname{End}_{U(3)}(V^{\otimes 6}) = \bigoplus_\eta I_{S_\eta(V)} \otimes \operatorname{End}(F_\eta).$$

The trace pairing identifies this commutant with simultaneous-unitary invariant multilinear polynomials. Its η block, denoted W_η, is exactly the direct sum over equivalent labels in source equation (12), with all six individual partitions equal to (1). In particular, $q_\eta \in W_\eta$. We use the whole block: we do not assume that q_η is one of the source's individually normalized Schur basis vectors.

3. All strictly lower terms are accounted for

Let L be the span of source Schur polynomials whose tuple of individual partitions is strictly below $((1), \dots, (1))$ in the source's order on pages 19–20. At the last differing position j, the partition must be zero. Thus each such polynomial is independent of X_j, even if it has arbitrarily high degree in earlier variables. Every polynomial multilinear in $Z_1, \dots, Z_6$ has conditional mean zero in each variable, and is therefore orthogonal to *all* of L.

Let $Cq = q(X_1 - I/2, \dots, X_6 - I/2)$ for a multilinear q. Then $Cq - q \in L$ and $Cq \perp L$; in particular $Cq_\eta = f_\eta$. The only length-at-most-three partitions of six strictly below either λ or μ are

$$\nu = (3, 2, 1), \qquad \omega = (2, 2, 2).$$

Indeed, all such partitions are $(6), (5,1), (4,2), (4,1,1), (3,3), (3,2,1), (2,2,2)$, and comparison of their partial sums proves the assertion. Consequently the full strictly lower space for either target block is the orthogonal sum

$$L \oplus N, \qquad N = C(W_\nu \oplus W_\omega). \tag{3}$$

This identity uses $w - Cw \in L$ for each w in the two lower blocks. It does not require a finite-dimensional assumption about L.

The product measure is invariant under permutation of the six slots. The finite-dimensional space N has the same invariance. In a block W_η, the slot action is conjugation by the irreducible S_6 representation on F_η. By Schur's lemma its fixed space consists precisely of scalar multiples of the identity of F_η. Thus the permutation-fixed subspace of N is

$$N^{S_6} = \operatorname{span}\{f_\nu, f_\omega\}.$$

Orthogonal projection onto N commutes with slot permutations. Since f_λ and f_μ are fixed, their projections belong to this two-dimensional subspace. Projecting away these two vectors therefore removes *every* strictly lower term required by the source, including all multiplicity copies.

For precision, the map taking a leading polynomial $q \in W_\eta$ to the polynomial with that leading term, lower corrections, and orthogonality to the lower space is linear and unique. Existence here follows by first applying C and then projecting away N. For uniqueness, the difference of two possible answers is lower and orthogonal to itself, hence zero because the measure has positive density on an open set. Applying this linear map to a Schur basis of W_η gives exactly Definition 2.23. Thus its value on q_η is a linear combination of the source's Jacobi polynomials with final label η, for *any* allowed choice of basis.

4. Exact Gram calculation

For $r \geq 0$, let $f^{\eta/\xi}$ denote the number of standard tableaux of skew shape η/ξ, taken as zero unless $\xi \subseteq \eta$. Write $f^\xi = f^{\xi/\varnothing}$ and $D_\varnothing = 1$. The partial trace of E_η over $6 - r$ slots is

$$\operatorname{Tr}_{6-r} E_\eta = \sum_{\xi \vdash r,\, \ell(\xi) \leq 3} \frac{D_\eta}{D_\xi} f^{\eta/\xi} E_\xi^{(r)}. \tag{4}$$

To see this, it commutes with both $U(3)$ and S_r, so is scalar on each ξ summand. The branching rule from S_6 to S_r gives

$$\operatorname{Tr}\big(E_\eta (E_\xi^{(r)} \otimes I)\big) = D_\eta f^\xi f^{\eta/\xi}.$$

Dividing by $\operatorname{Tr} E_\xi^{(r)} = D_\xi f^\xi$ proves (4).

Independence and (1) give the following normalized Gram matrix:

$$G_{\eta\zeta} := a^{-6}\langle f_\eta, f_\zeta \rangle = \sum_{r=0}^{6} \binom{6}{r} \left(-\frac{1}{6}\right)^{6-r} \sum_{\xi \vdash r,\, \ell(\xi) \leq 3} \frac{f^\xi}{D_\xi} f^{\eta/\xi} f^{\zeta/\xi}. \tag{5}$$

Indeed, choosing the a term of the covariance in r slots and the b term in the others contracts the two partial traces in (4). The trace of their product, divided by $D_\eta D_\zeta$, is the inner sum in (5). Slot symmetry gives the binomial coefficient.

In the order $(\lambda, \mu, \nu, \omega)$, evaluation gives

$$G = \frac{1}{46656} \begin{pmatrix} 15604 & 2510 & -14576 & 11330 \\ 2510 & 14647 & -4480 & 3505 \\ -14576 & -4480 & 29152 & -42640 \\ 11330 & 3505 & -42640 & 174505 \end{pmatrix}. \tag{6}$$

All entries can be calculated using only the corner-removal recurrence

$$f^{\eta/\xi} = \sum_{\substack{\eta^- \text{ obtained by removing a corner} \\ \xi \subseteq \eta^-}} f^{\eta^-/\xi}, \qquad f^{\xi/\xi} = 1,$$

and the dimension formula

$$D_\xi = \prod_{1 \le i < j \le 3} \frac{\xi_i - \xi_j + j - i}{j - i},$$

with zero parts appended. The accompanying exact verifier implements these formulas. It also verifies all sixteen entries independently, by constructing the four 729×729 integer matrices $\sum_{\sigma \in S_6} \chi^\eta(\sigma) T_\sigma$ directly and taking their partial traces. In that calculation characters are computed as the coefficient of $x^{\eta+(2,1,0)}$ in

$$\prod_{i<j}(x_i - x_j) \prod_{h \in \mathrm{cycles}(\sigma)} (x_1^{|h|} + x_2^{|h|} + x_3^{|h|}),$$

without using the skew-tableau recurrence. Every operation is exact.

5. The nonzero residual pairing

Partition G into target indices $A = (\lambda, \mu)$ and lower indices $B = (\nu, \omega)$. The matrix G_{BB} is positive definite, and

$$G_{AB} G_{BB}^{-1} = -\frac{1}{1513426} \begin{pmatrix} 953923 & 134828 \\ 292745 & 41134 \end{pmatrix}.$$

Thus the required polynomials are

$$p_\lambda = f_\lambda + \frac{953923 f_\nu + 134828 f_\omega}{1513426},$$
$$p_\mu = f_\mu + \frac{292745 f_\nu + 41134 f_\omega}{1513426}.$$

By Sections 2–3 these lie in the claimed Jacobi spans and are orthogonal to their full strictly lower spaces. Their normalized Gram matrix is the Schur complement

$$a^{-6} \begin{pmatrix} \langle p_\lambda, p_\lambda \rangle & \langle p_\lambda, p_\mu \rangle \\ \langle p_\mu, p_\lambda \rangle & \langle p_\mu, p_\mu \rangle \end{pmatrix} = G_{AA} - G_{AB} G_{BB}^{-1} G_{BA} = \frac{1}{24214816} \begin{pmatrix} 3854156 & -790 \\ -790 & 7201587 \end{pmatrix}.$$

The off-diagonal entry proves the theorem. If every pair of the source's basis polynomials from the two distinct blocks were orthogonal, every linear combination from the first would be orthogonal to every linear combination from the second. The displayed strictly negative number rules this out. The argument therefore does not exploit the source's exception for equivalent indices. □

Task 2. Locate the smallest examples

Determine the least number of matrix variables allowing failure at $m = 3$ and total degree six. In particular, investigate two variables with $\tau_1 = \tau_2 = (2, 1)$, whose product contains both target partition types. For $m \le 2$, the source's partition order is total, so distinct non-equivalent indices are already comparable; this identifies a natural boundary for the search.

Task 3. Classify the parameter dependence

For six identically distributed matrix beta variables, replace the ratio $b/a = -1/6$ by its general value. Determine the exact rational function obtained from (5) and the same Schur complement, its zeros, and its sign throughout the admissible parameter region. This should quantify how the obstruction persists under perturbation.

Task 4. Give individually labelled witnesses

Fix explicit standard-tableau matrix units in each multiplicity space and compute a particular pair of original Jacobi basis polynomials with nonzero pairing. Compare its coefficients and size with the basis-independent central certificate given here.

Task 5. Formulate a corrected orthogonality statement

Determine which incomparable partition blocks must be grouped together to admit an orthogonal construction preserving the original individual matrix degrees. Study whether additional symmetric operators can separate the resulting blocks, and compare this with Gram–Schmidt for a total extension of dominance order.

References

[1] T. Kuijper, *Jacobi polynomials on the simplex of Hermitian matrices and zonal spherical functions on partial flag manifolds*, arXiv:2607.25760v1 (2026), equations (7), (8), (12), Definition 2.23 and Conjecture 2.25.

Verification and status. The accompanying Python verifier reproduces the finite arithmetic certificate (SymPy and NumPy required). The full current 40-page source retains the conjecture; dated claim checks accompany this note. Mathematical writing and review were performed by the authoring agent, without independent peer review.